Additional Problems for Ch 9 HRW – Centre of Mass and Momentum
Addisionele Probleme vir Hfst 9 HRW – Swaartepunt en Momentum

	9^{th} Edition	8^{th} Edition
Questions	3, 5	5, 3
Problems	2, 5, 22, 25, 38, 44, 52, 60, 68, 85, 89, 91, 97	2, 3, 20, 27, 36, 46, 53, 62, 68, 81, 93, 99, 103

2	2	(a) 1.1 m (b) 1.4 m (c) As the mass of m_3, the topmost particle, is increased, the center of mass shifts toward that particle. As we approach the limit where m_3 is infinitely more massive than the others, the center of mass becomes infinitesimally close to the position of m_3.	(a) 1.1 m (b) 1.3 m (c) As the mass of m_3, the topmost particle, is increased, the center of mass shifts toward that particle. As we approach the limit where m_3 is infinitely more massive than the others, the center of mass becomes infinitesimally close to the position of m_3.
5	3	(a) – 0.45 cm . (b) – 2.0 cm .	
22	20	(a) $\theta_2 = 30.0°$. (b) $(-0.520\ \text{kg}\cdot\text{m/s})\hat{\text{j}}$.	(a) $\theta_2 = 30.0°$. (b) $(-0.572\ \text{kg}\cdot\text{m/s})\hat{\text{j}}$.
25	27	(a) 42 kg.m/s (b) 2.1 x 10^3 N	
38	36	(a) $\vec{J} = (2.4\ \text{N}\cdot\text{s})\hat{\text{j}}$ (b) The force on the ball by the wall is $\vec{J}/\Delta t = (2.4/0.010)\hat{\text{j}} = (240\,\text{N})\,\hat{\text{j}}$. By Newton's third law, the force on the wall by the ball is $(-240\ \text{N})\hat{\text{j}}$ (that is, its magnitude is 240 N and its direction is directly into the wall, or "down" in the view provided by Fig. 9-54).	(a) $\vec{J} = (1.8\ \text{N}\cdot\text{s})\hat{\text{j}}$ (b) The force on the ball by the wall is $\vec{J}/\Delta t = (1.8/0.010)\hat{\text{j}} = (180\,\text{N})\,\hat{\text{j}}$. By Newton's third law, the force on the wall by the ball is $(-180\ \text{N})\hat{\text{j}}$ (that is, its magnitude is 180 N and its direction is directly into the wall, or "down" in the view provided by Fig. 9-55).
44	46	$m_R = 1.29$ kg and $m_R + m_L \approx 3.3$ kg.	$m_R + m_L \approx 3.4$ kg.
52	53	$h = 0.10$ m.	$h = 0.073$ m.
60	62	(a) $v_{af} = 1.9$ m/s (b) The block continues going to the right after the collision. (c) Since $K_i = K_f$ the collision is found to be elastic.	
68	68	$h = 2.50$ m, $\mu_k = 0.500$ (a) $d = 2.67$ m. (b) $d = 0.666$	$h = 2.50$ m, $\mu_k = 0.500$ (a) $d = 2.22$ m. (b) $d = 0.556$ m

85	81	(a) $v_{3ff} \approx 1.78$ m/s. (b) We see that v_{3ff} is less than v_{1i} . (c) $K_{3ff} = \frac{64}{81} K_{1i}$. We see that this is less than K_{1i} . (d) The final momentum of block 3 is $p_{3ff} = m_3 v_{3ff} = (4m_1)\left(\frac{16}{9}\right)v_1 > m_1 v_1$.
89	93	(a) $\vec{J} = \vec{p}_f - \vec{p}_i = \left(7.4\times10^3\ \mathrm{N\cdot s}\right)\left(\hat{\mathrm{i}} - \hat{\mathrm{j}}\right)$. (b) $\vec{J} = \vec{p}_f - \vec{p}_i = (-7.4\times10^3\,\mathrm{N\cdot s})\hat{\mathrm{i}}$. (c) $F_{\text{avg}} = (1600\,\mathrm{N})\sqrt{2} = 2.3\times10^3\ \mathrm{N}$. (d) $F_{\text{avg}} = 2.1 \times 10^4$ N. (e) The average force is given above in unit vector notation. Its *x* and *y* components have equal magnitudes. The *x* component is positive and the *y* component is negative, so the force is 45° below the positive *x* axis.
91	99	+ 4.4 m/s.
97	103	(a) 6.9 m/s. (b) The direction of ball 2 is at 30° counterclockwise from the +*x* axis. (c) 6.9 m/s. (d) The direction of ball 3 is at −30° counterclockwise from the +*x* axis. (e) Now we use the momentum equation to find the final velocity of ball 1: $V = v_0 - 2v\cos\theta = 10\ \ \mathrm{m/s} - 2(6.93\ \ \mathrm{m/s})\ \cos 30° = -2.0\,\mathrm{m/s}$.